

Solution Architecture

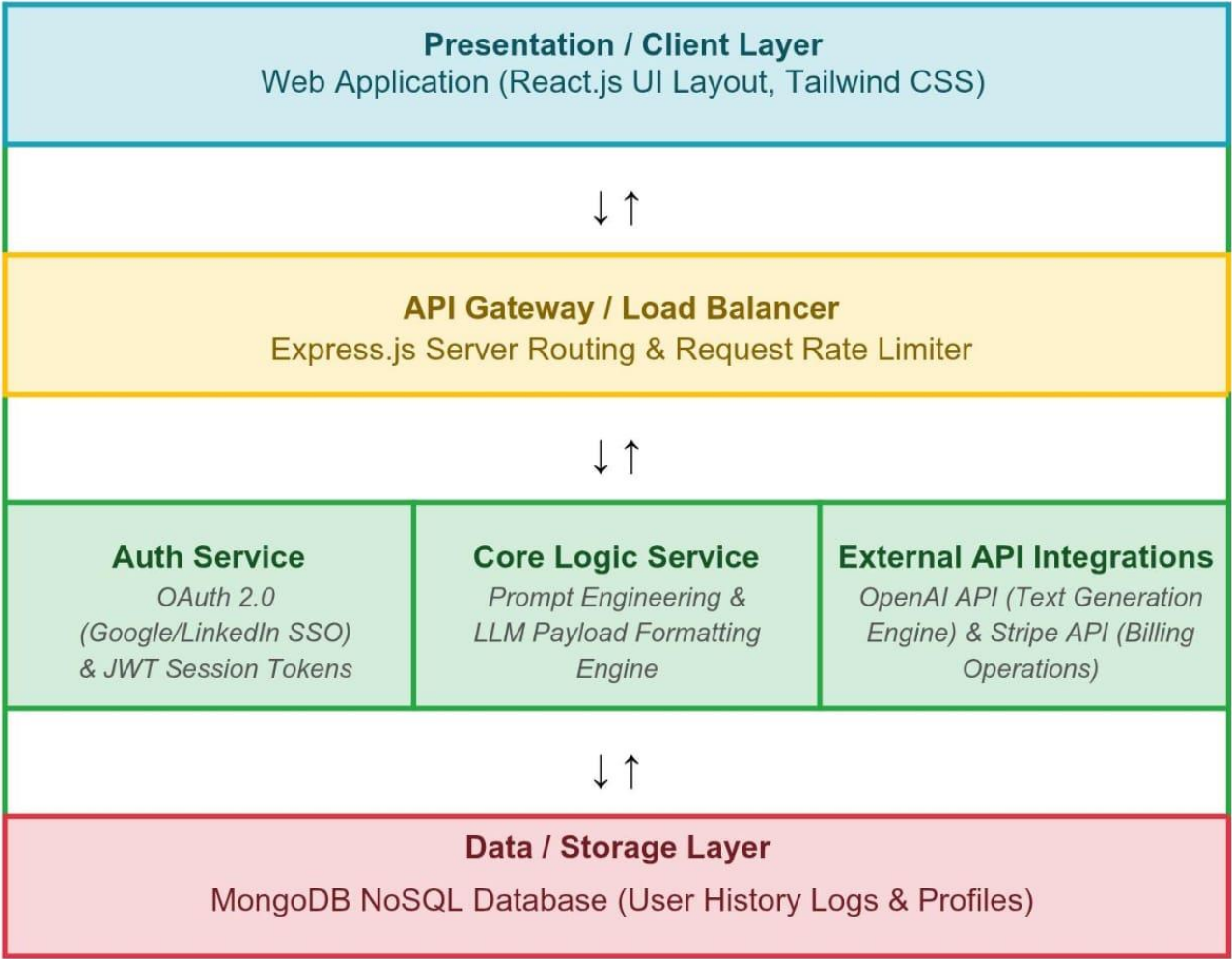
Project Name: CaptionAI: AI-powered social media caption generator

Date: 7 July 2026

Team Name: Anupriya

Maximum Marks: 5 Marks

1. Solution Architecture Diagram Mapping



2. Component Description Table

Component Name PDF	Description / Role in Architecture PDF	Technologies Used PDF
Presentation Layer	Manages user interaction, displays custom input forms for product text/platform preferences, and renders final platform-optimized copies with interactive single-click copy containers.	React.js, Tailwind CSS
API Gateway	Validates client-side authentication tokens, acts as a single entry point for incoming user payloads, and manages safe system traffic routing.	Express.js, JWT (JSON Web Tokens)
Microservices / Core Logic	Processes incoming product descriptions, shapes them into platform-specific prompt templates, communicates with generative text endpoints, and logs active usage parameters.	Node.js, OpenAI API
Database	Persists application metadata, manages user account credentials safely, and stores historically generated social media caption data for convenient dashboard reference.	MongoDB, Mongoose ODM